EFY TESTED
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High-Fidelity FM TRANSMITTER

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FM broadcasting is radio broadcasting using frequency modulation technology. In order to attain high-fidelity broadcast of music and speech, all audio frequencies up to 15kHz must be transmitted. The low-power transmitter presented here achieves that using simple and readily available components. Its output frequency is locked to 96MHz using the phase-locked loop (PLL) approach.

PARTS LIST

Semiconductors:	
IC1	- 74AC74 dual D-type flip-flop with preset and clear
IC2	- 7805, 5V regulator
IC3, IC5	- CD4060 ripple-carry binary counter
IC4	- CD4046 phase-locked loop
IC6	- LM386 low-power audio amplifier
LED1	- 5mm LED
D1	- BB910 varactor diode
T1, T3	- PN2222A npn transistor
T2	- 2N3904 npn transistor
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2, R10	- 22-kilo-ohm
R3	- 220-ohm
R4, R5, R6, R9	- 10-kilo-ohm
R7, R12, R13	- 2.2-kilo-ohm
R8, R11, R17, R19	- 100-kilo-ohm
R14	- 47-kilo-ohm
R15, R16	- 33-kilo-ohm
R18	- 330-ohm
VR1	- 470-kilo-ohm potmeter
VR2	- 10-kilo-ohm potmeter
Capacitors:	
C1, C7, C9-C11, C14, C23	- 100nF ceramic disk
C2	- 330pF ceramic disk
C3	- 10pF ceramic disk
C4	- 2.2pF ceramic disk
C5	- 3pF ceramic disk
C6, C8	- 100µF, 25V electrolytic
C12, C20	- 220µF, 25V electrolytic
C13	- 22µF, 25V electrolytic
C15-C17	- 10µF, 25V electrolytic
C18	- 2.2µF, 25V electrolytic
C19	- 1000pF ceramic disk
C21, C22	- 22pF ceramic disk
VC1	- 50pF trimmer
Miscellaneous:	
RFC1	- 100µH, 2-hole balun ferrite core
RFC2, RFC3	- 330µH readymade axial inductor
L1	- 5T 20 SWG 8mm dia. air core spread over 1.2 or 1.5cm. Tap at 1/2T from V+
JACK1, JACK2	- Audio input jacks
XTAL1	- 12MHz crystal oscillator
BATT.1	- 12V battery or 12V DC power supply
ANT.1	- 78cm single-strand wire as antenna

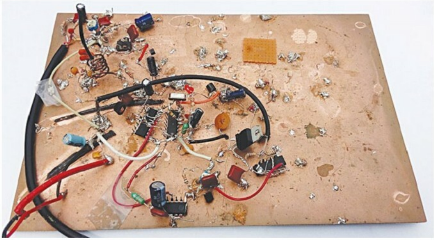

Fig. 1: Author's prototype wired on a copper clad sheet

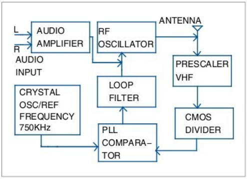

Fig. 2: Block diagram of the high-fidelity FM transmitter

Test Points

Test point	Details
TP0	0V (GND)
TP1	12V
TP2	5V
TP3	Amplified audio signal
TP4	3.6V-4.2V
RF OUT	96MHz
LED1	'On' when frequency is locked

Circuit and working

The author's prototype of the high-fidelity FM transmitter circuit wired on a copper clad sheet is shown in Fig. 1. Block diagram and circuit are shown in Figs 2 and 3, respectively. The circuit is built around dual D-type positive-edge-triggered flip-flop 74AC74 (IC1), ripple-carry binary counter CD4060 (IC3 and IC5), low-power audio amplifier LM386 (IC6), phase-locked loop CD4046 (IC4), 5V voltage regulator 7805 (IC2) and a few other components.

Transistor T1 along with resistors R1, R2 and R3, capacitors C2 and C5, coil L1, trimmer capacitor VC1, and varactor diode D1 forms a voltage-controlled oscillator (VCO) in Pierce configuration. IC1 acts as a VHF prescaler and divides the oscillator frequency by a factor of 4. Low-level RF output is taken via half-turn tap off L1 and fed to pin 3 of IC1 via C3 and to antenna via C4. The 24MHz (96MHz/4) output from the prescaler is further divided by 32 by binary counter IC3. This